

# EV Market Intelligence - Data Quality Report

Scope: India, 2019-2030, state/UT + city layer, quarterly EV sales and charging, annual demographics/economics/environment.

## Table Counts

| Table                        | Rows  | Columns |
|------------------------------|-------|---------|
| dim_regions                  | 270   | 9       |
| fact_ev_sales                | 12960 | 13      |
| fact_charging_infrastructure | 12960 | 14      |
| dim_demographics             | 3240  | 9       |
| fact_economic_data           | 3240  | 11      |
| dim_policies                 | 141   | 10      |
| fact_competitor_presence     | 2430  | 7       |
| fact_environmental_data      | 3240  | 8       |
| Combined                     | 38481 |         |

## Validation Checks

| Check                  | Result        |
|------------------------|---------------|
| Primary keys unique    | Pass          |
| Foreign key orphans    | Pass          |
| Negative sales         | Pass          |
| Negative population    | Pass          |
| Market share > 100     | Pass          |
| EV penetration min/max | 0.48% / 9.25% |
| AQI min/max            | 25.0 / 120.2  |
| Minimum charger count  | 1             |

## Notes

- The dataset is synthetic but uses business rules to keep values realistic across states, union territories, and city tiers.

- Southern and western states generally show stronger EV adoption, stronger infrastructure readiness, and broader brand coverage.
- Policies, charging infrastructure, and competitor shares were intentionally correlated with adoption and economic readiness.
- The structure is suitable for Excel analysis, Power BI modeling, and loading into MySQL/PostgreSQL.

## **Source inspiration used for the schema design**

- Kaggle - Electric Vehicle Charging Stations in India: <https://www.kaggle.com/datasets/saketpradhan/electric-vehicle-charging-stations-in-india>
- Kaggle - EV Charging Stations in India Simplified (2025): <https://www.kaggle.com/datasets/pranjal9091/ev-charging-stations-in-india-simplified-2025>
- Kaggle - Indian Electric Vehicle Dataset: <https://www.kaggle.com/datasets/sanhitasaxena/indian-electric-vehicle-dataset>
- Kaggle - Electric Vehicle Charging Station Usage Dataset: <https://www.kaggle.com/datasets/prince7489/electric-vehicle-charging-station-usage-dataset>
- Kaggle - Global EV Charging Stations Dataset: <https://www.kaggle.com/datasets/vivekatrri/global-ev-charging-stations-dataset>